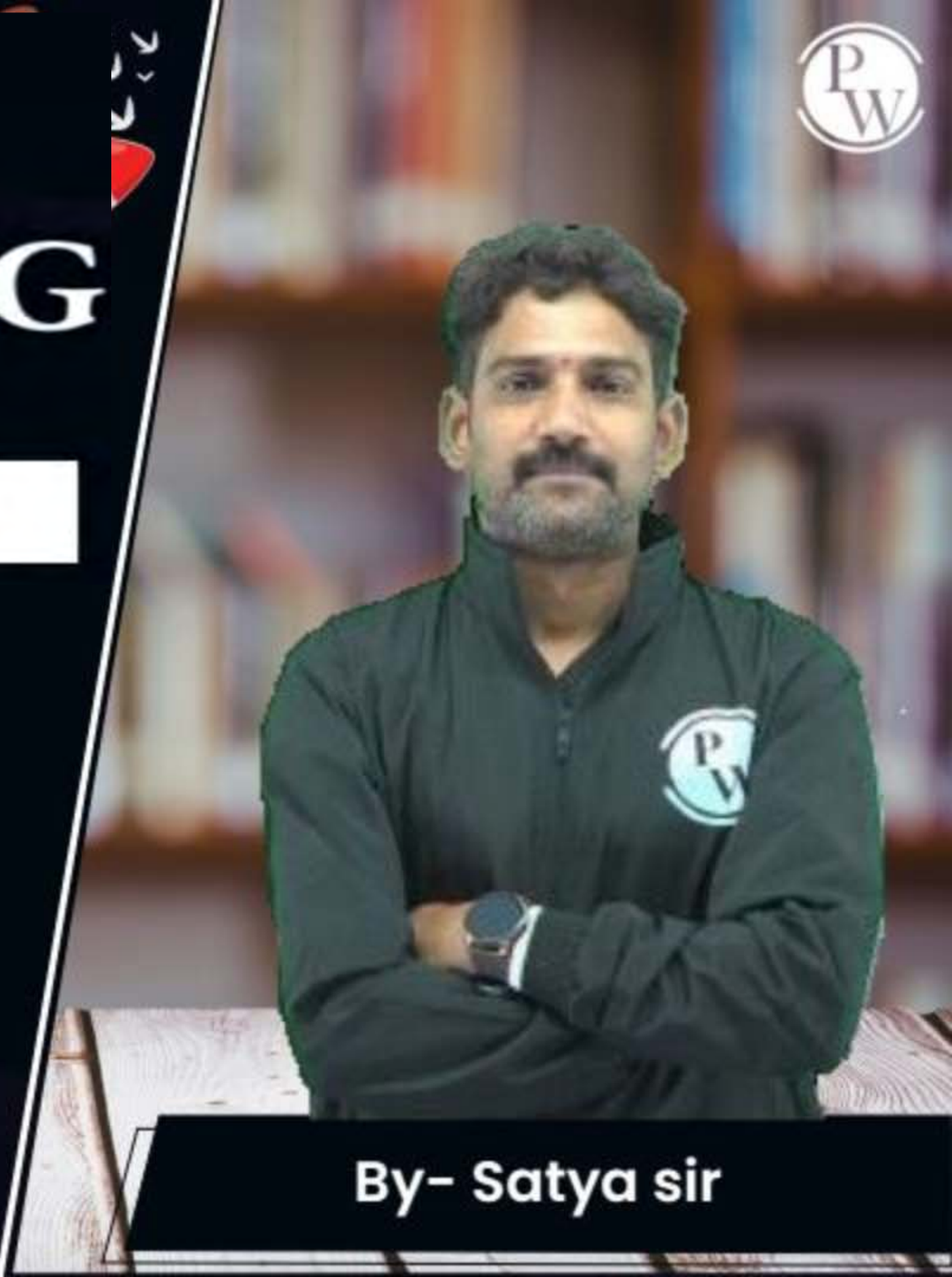
CS & IT ENGINEERING

Python For Data Science

Control Statements and Functions and Methods

DPP 01 (Discussion Notes)

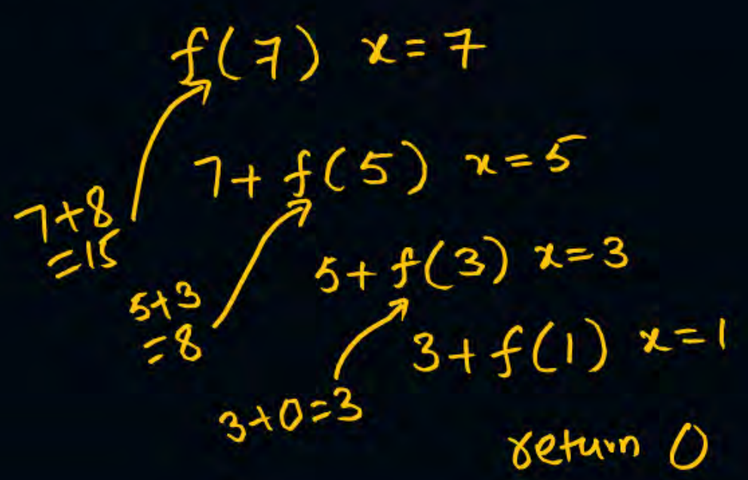


By- Satya sir

[NAT]

#Q. The return value of below function f(7) is 15

```
def f(x):
    if x<=1:
        return x-1
    return x + f(x-2)
```



[MCQ]

#Q. The output printed by below code is _____

```
def fun(i, j):
    if i==j:
        print(i+j, end=" ")
    else:
        print(i-1, j, end=" ")
        fun(i-2, j + 2)
→ fun(12, 0)
```

i j
fun(12, 0)
→ Print 11 0
fun(10, 2)
→ Print 9 2
fun(8, 4)
→ Print 7 4
fun(6, 6)
→ Print 12

o/p: 11 0 9 2 7 4 12

A 12 0 10 2 8 4 12

B 11 1 9 3 7 5 12

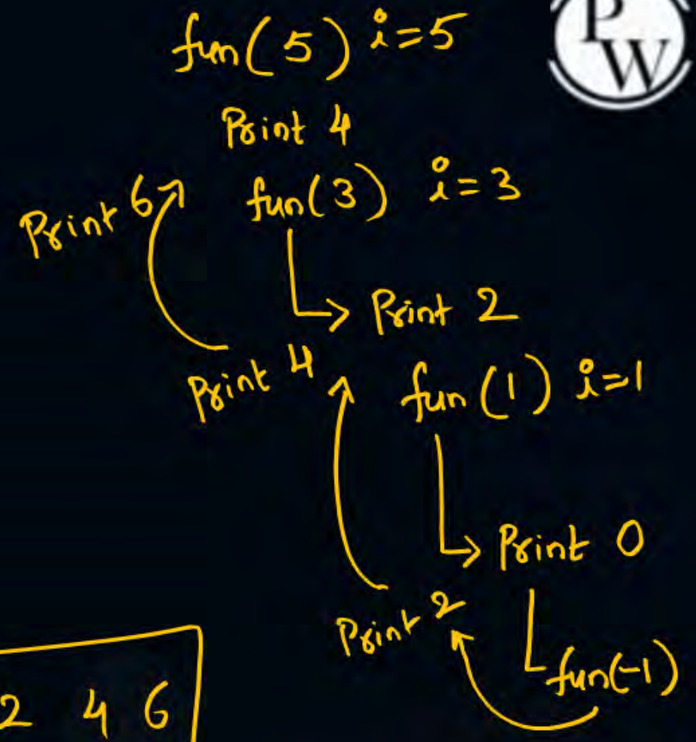
C ✓ 11 0 9 2 7 4 12

D 12 1 10 3 8 5 12

[MCQ]

#Q. The output printed by below code segment is _____

```
def function(i):
    if i<=0:
        return 0
    print(i -1,end=' ')
    function(i - 2)
    print(i+1, end=' ')
function(5)
```



o/p: 4 2 0 2 4 6

A 5 3 1 3 5 7

B 4 2 0 2 4 6

C 5 3 1 2 4 6

D 5 3 1 1 3 5

[NAT]

#Q. Consider the below code:

```
def f(i):
    count =1
    if i <= 0:
        return
    for x in range(i):
        k = count+g(x + 1)
        count = count + x
    return k
def g(i):
    j =1
    if i < 1:
        return i + 1
    for x in range(i + 1):
        j = j + x
    return j
```

The return value of f(5) is _____

23

Count=1

f(5) i=5

x=0 k = 1 + g(1) = 1+2 = 3, Count = 1+0 = 1

x=1 k = 1 + g(2) = 1+4 = 5, Count = 1+1 = 2

x=2 k = 2 + g(3) = 2+7 = 9, Count = 2+2 = 4

x=3 k = 4 + g(4) = 4+11 = 15, Count = 4+3 = 7

x=4 k = 7 + g(5) = 7+16 = 23, Count = 7+4 = 11

g(1) i=1

j=1

x=0 j=1+0=1

x=1 j=1+1=2

return 2

g(2) i=2

j=1

x=0 j=1

x=1 j=2

x=2 j=4

return 4

g(3) i=3

j=1

x=0 j=1

x=1 j=2

x=2 j=4

x=3 j=7

return 7

g(4) i=4

j=1

x=0 j=1

x=1 j=2

x=2 j=4

x=3 j=7

x=4 j=11

return 11

g(5) i=5

j=1

x=0 j=1

x=1 j=2

x=2 j=4

x=3 j=7

x=4 j=11

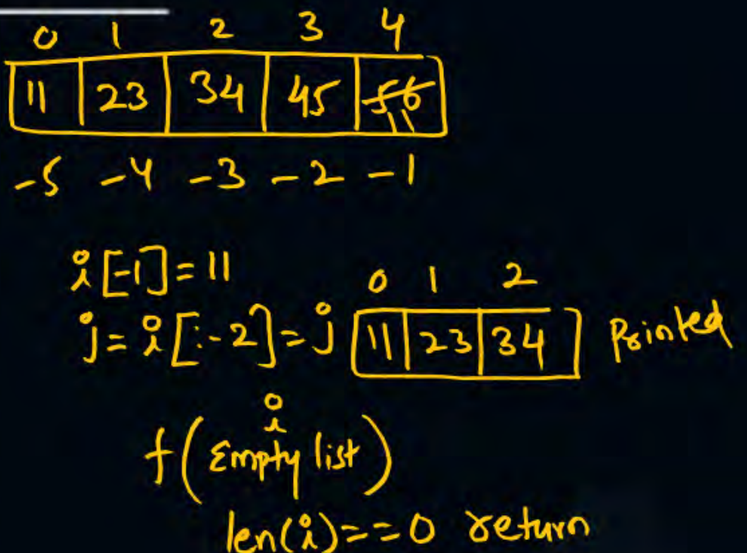
x=5 j=16

return 16

[MCQ]

#Q. The output printed by below code is _____

```
def f(i):
 if len(i)==0:
    return
 else:
    i[– 1]= i[0]
    j=i[: – 2]
    print(j)
    f(j[3:])
a=[11, 23, 34, 45, 56]
f(a)
```



A [11,23]

B [11,23,34] ✓

C [11,23,34,45]

D [11,23,34,45,56]

[NAT]

#Q.
```
def fun(n):
    x = 1
    if n == 1:
        return x
    for k in range(1, n):
        x += fun(k) * fun(n – k)
    return x
```

The return value of above code on fun(3) call is 5

fun(3) n=3

x=1

k=1 x=1+fun(1)*fun(2) ⇒ x=1+1*2=3

k=2 x=3+fun(2)*fun(1) ⇒ x=3+2*1=5

return 5

fun(1) n=1

x=1 return 1

fun(2) n=2

x=1

k=1 x=1+fun(1)*fun(1)

=1+1*1

=2

return 2

[MCQ]

#Q. The output printed is _____

```
def foo(n: int, r: int):
 if n > 0:
    return (n % r) + foo(n // r, r)
 else:
    return 0
print(foo(513, 2))
```

foo(513, 2)
1 + foo(256, 2)
0 + foo(128, 2)
0 + foo(64, 2)
0 + foo(32, 2)
0 + foo(16, 2)
0 + foo(8, 2)
0 + foo(4, 2)
0 + foo(2, 2)
0 + foo(1, 2)
1 + foo(0, 2)
0

1 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 1 + 0
= 2

A 0

B 1

C 2

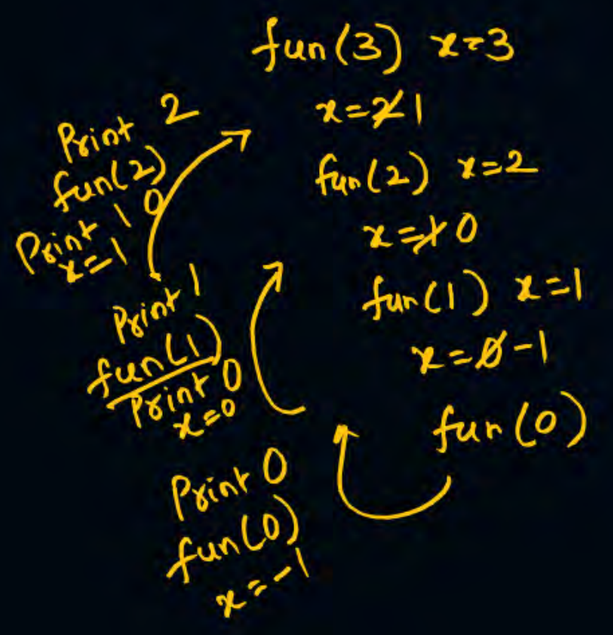
D 9

[MCQ]

#Q. The output of below python code segment is _____

```
def fun(x):
 if x > 0:
    x = x - 1
    fun(x)
    print(x, end='')
    fun(x)
    x = x - 1
fun(3)
```

o/p: 0 1 0 2 0 1 0



fun(1) Print 0

fun(2) Print 0 1 0

A 0102010

B 2012010

C 1010202

D 0101010

THANK - YOU